

An Enhanced Smith-Waterman Algorithm Using Adaptive Banding and Aware Gap Penalty Optimized by Starfish Optimizer

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Abstract—The rapid growth of genomic information makes scalable sequence alignment challenging. Although the Smith-Waterman method guarantees mathematical accuracy in aligning nucleotide sequences, its time complexity becomes a hindrance. This paper proposes a hybrid technique aimed at optimizing computation without sacrificing alignment sensitivity. Specifically, the algorithm combines k-mer-based similarity measurement with a positioning indexing system to construct an approximate search boundary. In addition, the Starfish Optimization Algorithm (SFOA) is utilized to refine the boundary by exploring the search space and exploiting the optimal path. Ultimately, the improved Smith-Waterman operation is implemented within this optimized search corridor with consideration of context-aware gap penalties. The experimental results show that the proposed strategy not only improves the alignment scores compared to classical banded algorithms, but also greatly reduces computational costs in terms of CPU runtime and Dynamic Programming (DP) cells.

Index Terms—Sequence Alignment, Smith-Waterman Algorithm, Banded Programming, Adaptive Banding, Starfish Optimization, Context-Aware Gap Penalty

I. INTRODUCTION

The latest breakthroughs in the Third-Generation Sequencing (TGS) technology of Pacific Biosciences and Oxford

Nanopore have made it possible to produce very long sequences of DNA (greater than 10^5 base pairs) [1]. Long reads are essential in the detection of complex structural variations and the study of new populations; however, they possess significant stochastic errors and an increasing search space as their lengths increase. The Smith-Waterman algorithm ensures the production of the best results for local sequence alignment and is still considered the gold standard for sequence alignment. However, it suffers from quadratic time and space complexity $O(mn)$ when handling large sequences, making it a fundamental bottleneck in large-scale genomic applications despite its mathematical superiority [2].

Traditional heuristic optimizers, including static banding or seed-and-extend approaches such as BLAST [3], usually face the difficulty of providing the optimal balance between sensitivity and efficiency. In addition to these heuristic techniques, there are some optimized methods that provide exact alignment based on a bound-and-align strategy, such as QuickEd [4]; nevertheless, these approaches are less suitable when dealing with sequences with very high mutation rates.

This study proposes the usage of *Adaptive Banding*, where the alignment corridor adapts its width based on the sequence

entropy and similarity to limit redundant computation while preserving the optimal path. Besides, traditional methods usually use constant gap penalties independent of evolutionary distance, so this paper proposes *context-aware gap penalty* adjustment depending on the sequence to increase the biological accuracy of alignment. As an efficient optimization method, bio-inspired metaheuristic algorithms can be applied since they provide advantages in finding solutions in non-convex similarity landscapes [5]. In particular, the Starfish Optimization Algorithm (SFOA) [6] is chosen for its unique multi-arm searching capability.

The main contribution of this research is as follows.

- Introducing an adaptive corridor width derived from an inverse function of K-mer/Jaccard similarity to dynamically adjust search boundaries.
- Proposing a context-aware gap penalty that dynamically adjusts penalties based on local mutation density, allowing divergent regions to be treated differently from highly conserved regions.
- Integrating the Starfish Optimization Algorithm (SFOA) to refine the initial band estimate, balancing exploration and exploitation to converge on a near-optimal corridor width.
- Designing an integrated alignment pipeline that combines k -mer prefiltering, SFOA-based corridor refinement, and context-aware gap penalties within a single restricted dynamic-programming pass, avoiding full-matrix computation.
- Conducting an empirical evaluation on real biological datasets (human mitochondrial DNA and *E. coli* genomes) that benchmarks alignment score, execution time, and computed DP cells against standard Smith–Waterman, fixed banding, and a BLAST-like heuristic.

II. RELATED WORK

A. Classical Local Alignment and Heuristic Approaches

Sequence alignment sits at the heart of bioinformatics, powering everything from genome analysis to mutation detection and studies of evolutionary history. Smith and Waterman [7] introduced a classic method known for producing the best possible local alignments using Dynamic Programming. However, as detailed by Edi et al. [1] and Xia et al. [8], its quadratic complexity makes it too slow for massive datasets generated by Oxford Nanopore and Pacific Biosciences technologies.

This bottleneck is mitigated by heuristic approaches that prioritize speed over exhaustive search. Altschul et al. [3] developed BLAST, which uses a seed-and-extension heuristic that greatly improves runtime. However, as noted by Mount [9], BLAST may miss weak biological matches due to mutations or structural divergence.

B. Space-Bounded and Adaptive Dynamic Programming

Constrained Dynamic Programming applies limits to the search space to minimize computation without sacrificing alignment sensitivity. As discussed by Doblas et al. [4], banded alignment narrows the search to a predefined diagonal

band, reducing complexity to approximately $O(nb)$ where b is the bandwidth. However, strict banding suffers from a rigid tradeoff: small bands may miss evolutionary paths while large ones render computation inefficient.

Several recent studies have explored adaptive variants of banded Dynamic Programming to overcome the limitations of fixed-width corridors [4], [10]. These methods dynamically adjust the search region according to sequence characteristics, providing a better balance between computational efficiency and alignment sensitivity than conventional static banding techniques [8], [11].

C. Alignment-Free Approaches and Indexing Optimization

Alignment-free approaches circumvent Dynamic Programming through statistical measures such as k -mer frequencies, sketches, and hash tables. *Mash*, created by Ondov et al. [12], performs rapid MinHash-based distance estimation. Shi et al. [10] performed genome clustering without references by analyzing k -mer frequencies. Li and Guidi [13] introduced a scalable distributed-memory k -mer counting system, while Al-Baik et al. [14] presented the *Pufferfish* indexing approach for lightweight pre-alignment filtering.

Recently, Charvel et al. [11] introduced ReSkmer, demonstrating that repeat-aware k -mer representations significantly improve the accuracy of alignment-free genomic distance estimation, particularly in population-scale genomic analyses. Their work underscores the effectiveness of k -mer-based similarity estimation as an efficient preprocessing strategy prior to executing computationally expensive alignment algorithms.

D. Metaheuristic Optimization and Gap Penalty Modeling

Most standard alignment algorithms apply fixed gap penalties, which fail to capture the variable mutation rates found in real biological sequences. Context-aware models that set gap penalties based on local entropy rates have been proposed to address this. As reviewed by Jakšić et al. [5], swarm-based and evolutionary algorithms are well-suited for optimizing such non-linear scoring functions. Recent frameworks including the SFOA [6] and the Pufferfish Optimization Algorithm [14] offer promising paths for high-dimensional global optimization. Furthermore, Nandi et al. [15] proposed a hybrid approach combining chaotic metaheuristic strategies with multiple sequence alignment to improve convergence quality while avoiding local optima. Their findings further support the integration of bio-inspired optimization techniques for enhancing alignment performance and search-space exploration.

III. PROBLEM FORMULATION

Exhaustive Smith–Waterman alignment [7] guarantees the optimal local alignment between two sequences, but its quadratic time and space complexity makes full-matrix computation impractical for the read lengths produced by contemporary long-read sequencing [1], [8]. Heuristic alternatives such as BLAST [3], [9] and fixed-width banded alignment [4] restore speed, but both impose a rigid, sequence-independent search strategy: BLAST can miss weak or divergent biological

matches, while static banding assumes mutations and indels are distributed uniformly along the sequence. This assumption fails for real genomic data, where mutation density varies locally and the true evolutionary path can drift outside a fixed corridor, forcing a choice between wasted computation from an unnecessarily wide band and alignment failure from a band too narrow to contain the optimal path.

Existing adaptive-banding and alignment-free strategies [4], [10], [11] use sequence similarity to inform the search region, but the band width is typically fixed through a single heuristic mapping rather than optimized against an explicit accuracy-cost trade-off, and the gap penalty is generally applied uniformly despite local variation in mutation density. The problem addressed here is therefore twofold: (i) deriving an initial search corridor that reflects true sequence similarity without invoking full Dynamic Programming, and (ii) refining that corridor and its gap penalty so the restricted alignment approaches exhaustive Smith–Waterman accuracy while computing only a fraction of its Dynamic Programming cells.

This work formulates the solution as a search-space optimization task: an initial band width is estimated from k -mer similarity, then refined by the Starfish Optimization Algorithm [5], [6] under a fitness criterion balancing accuracy and cost, together with a gap penalty that scales with local mutation density. It is hypothesized that this joint optimization retains alignment scores close to exhaustive Smith–Waterman while substantially reducing Dynamic Programming cells and execution time relative to standard SW, fixed banding, and BLAST-like heuristics. The methodology developed to test this hypothesis follows in the next section.

IV. METHODOLOGY

A. System Overview

The proposed framework integrates lightweight similarity estimation and search space pruning to optimize local genomic sequence alignment. To overcome the scalability limitations of the standard SW algorithm, the system implements a three-stage pipeline [7], [8]: (1) k -mer indexing to estimate inter-sequence similarity, (2) bio-inspired heuristic search-space narrowing [1], [8], and (3) restricted DP execution [10], [13]. An overview of the framework architecture is shown in Fig. 1.

B. K -mer Based Similarity Estimation

Rather than performing full DP comparisons, a set-theoretic k -mer analysis is implemented as a prefilter [10]. Similarity is estimated using Jaccard similarity, which is defined as the ratio of shared elements to the union of both sets [11]:

$$J(K_1, K_2) = \frac{|K_1 \cap K_2|}{|K_1 \cup K_2|} \quad (1)$$

where K_1 and K_2 are the sets of k -mers extracted from sequences S_1 and S_2 , respectively.

The k -mer Jaccard similarity is integrated as an alignment-free metric for rapid preprocessing [10], [11]. By indexing the reference sequence with a lightweight hash table that maps each distinct k -mer to its positions, the query sequence is

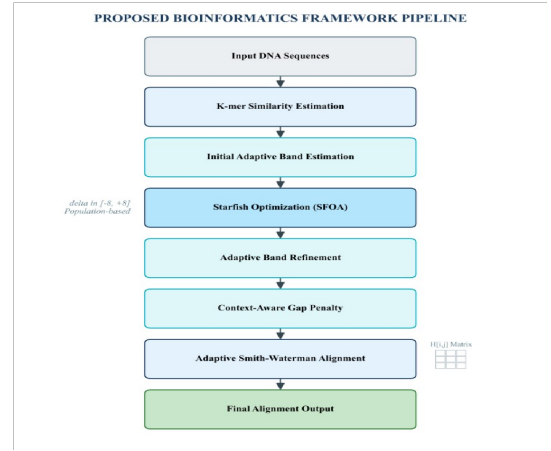


Fig. 1. Proposed enhanced Smith–Waterman framework architecture.

efficiently scanned. This allows the framework to directly compute the Jaccard similarity score J without resorting to computationally expensive Dynamic Programming [9], [11].

C. Adaptive Band Selection Strategy

In banded alignment, the SW algorithm is confined to a diagonal corridor of width b , reducing complexity from $O(mn)$ to $O(nb)$ [3], [8]. The challenge lies in selecting b : a very small value may miss the optimal path, while a large value negates computational savings [7].

The proposed system uses inverse mapping: the higher the sequence similarity, the smaller the required search band, and vice versa [9], [13]. This adaptive band width b is dynamically calculated using the following formulation:

$$b = b_{\max} \times (1 - J) + b_{\min} \quad (2)$$

where $b_{\max}$ and $b_{\min}$ represent the predefined upper and lower bounds of the search corridor, and J is the k -mer Jaccard similarity score obtained during the preprocessing stage. An illustration of this approach is shown in Fig. 2.

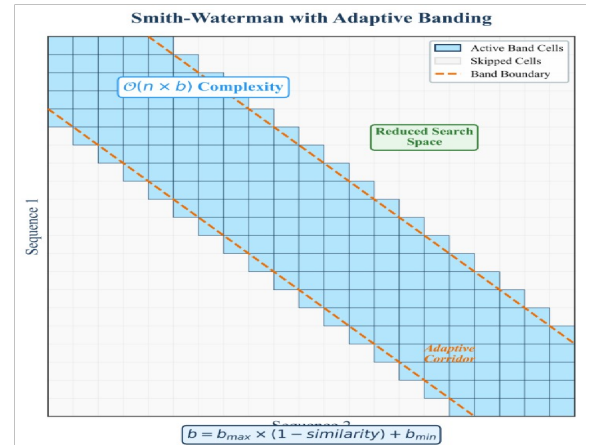


Fig. 2. Adaptive banded Smith–Waterman alignment.

D. Bio-Inspired Optimization

To further refine the search corridor beyond heuristic mappings, the Starfish Optimization Algorithm (SFOA) is used as a fine-tuner [6], balancing three processes as illustrated in Fig. 3:

- **Exploration:** Multi-arm environmental sensing to evaluate candidate band widths [6].
- **Exploitation:** Coordinated motion toward the global optimum using distance vectors from elite candidates, enabling fast convergence for highly similar pairs [6].
- **Regeneration:** Stochastic Lévy-flight perturbation of the worst-performing candidates, preventing entrapment in local optima during indel-rich segment alignment [6].

The SFOA search process determines the optimal corridor width b dynamically. The algorithm iteratively evaluates each candidate solution using a multi-objective fitness function designed to maximize alignment accuracy while minimizing computational overhead:

$$\text{Fitness}(b) = \alpha \cdot \frac{H_{\max}(b)}{H_{\max}^{\text{full}}} - \beta \cdot \frac{C(b)}{C_{\text{full}}} \quad (3)$$

where $H_{\max}(b)$ represents the maximum alignment score obtained within the band of width b , $H_{\max}^{\text{full}}$ is the optimal score of the full unconstrained matrix, $C(b)$ denotes the number of Dynamic Programming cells calculated inside the restricted band, and C_{full} is the total number of cells in the standard matrix. The weighting parameters α and β (where $\alpha + \beta = 1$) control the relative trade-off between alignment sensitivity and computational speed.

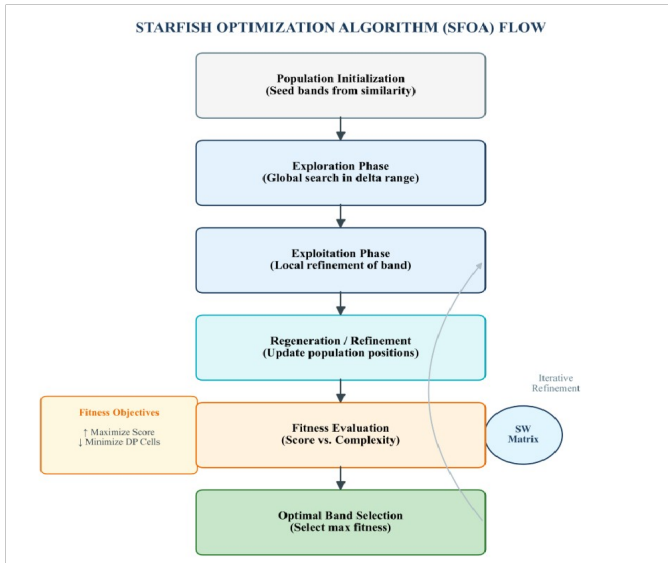


Fig. 3. Starfish Optimization Algorithm (SFOA) workflow.

E. Hybrid Alignment Pipeline

The integration process proceeds through five stages, as compared in Fig. 4:

- 1) **Pre-processing:** Sequences are translated into k-mer sets [13].
- 2) **Similarity Measurement:** k-mer Jaccard similarity $J(K1, K2)$ is computed [13].
- 3) **Band width Calculation:** The obtained Jaccard similarity is used to estimate the initial band width b_0
- 4) **Optimization:** SFOA refines the band to the optimal width b^* [6].
- 5) **Context-Sensitive Execution:** Adjusted SW alignment is run within b^* using context-specific gap penalties for local mutations [8].

Comparative Methodology — Matrix Computation Reduction

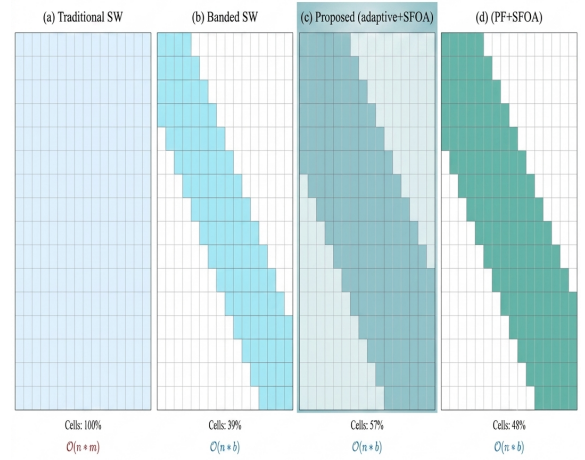


Fig. 4. Comparison between sequence alignment approaches.

F. Algorithm Pseudocode

V. RESULTS AND DISCUSSION

The specific parameter settings and experimental values used to evaluate the proposed framework are summarized in Table I.

TABLE I
SYSTEM PARAMETER SETTINGS AND EXPERIMENTAL VALUES

Parameter / Symbol	Value
Match Score (MATCH)	+2
Mismatch Penalty (MISMATCH)	-1
Baseline Gap Penalty (g_{base})	-2
alpha	0.7
beta	0.3
mutation window	10
k-mer Size (k)	6
Similarity Metric	Jaccard
Min Band Width ($b_{\min}$)	2
Max Band Width ($b_{\max}$)	50
Population Size (N)	20
Maximum Iterations (T)	30

Algorithm 1 Enhanced Smith–Waterman with Adaptive Banding and SFOA

Input: Genomic sequences (S_1, S_2) (FASTA format), weighting parameters (α, β), maximum number of iterations ($T_{\max}$), population size (N).

- 1: **// Phase 1: Preprocessing & Feature Extraction**
 - 2: Read and validate S_1 and S_2
 - 3: Compute k -mer similarity between S_1 and S_2 using Jaccard Similarity Using Eq. (1)
 - 4: Estimate the initial band width b_0 based on the computed Jaccard similarity: $b_0 = f(J)$
 - 5: **// Phase 2: Starfish Optimization (SFOA) Loop to Find b^***
 - 6: Initialize a population of starfish positions $X = \{b_1, b_2, \dots, b_N\}$ randomly within $[b_{\min}, b_{\max}]$
 - 7: Set iteration counter $t \leftarrow 0$
 - 8: **while** $t < T_{\max}$ **do**
 - 9: **for** each candidate starfish $b_i \in X$ **do**
 - 10: Compute Local Mutation Density $\mathcal{M}$ and Context-Aware Gap Penalty $W_c(k)$
 - 11: Run restricted Dynamic Programming within corridor of width b_i
 - 12: Evaluate $\text{Fitness}(b_i)$ using Eq. (3)
 - 13: **end for**
 - 14: Identify elite candidate b_{elite} with maximum fitness
 - 15: Update starfish positions (Exploration, Exploitation, and Lévy-flight Regeneration)
 - 16: $t \leftarrow t + 1$
 - 17: **end while**
 - 18: Select the global best solution as the optimized band width: $b^* \leftarrow b_{\text{elite}}$
 - 19: **// Phase 3: Alignment Execution and Benchmarking**
 - 20: Execute final adaptive Smith–Waterman alignment using optimized b^* and $W_c(k)$
 - 21: Track maximum alignment score $H_{\text{best}} \leftarrow H(S_1, S_2, b^*)$
 - 22: **return** H_{best} , b^* , and alignment statistics.
-

TABLE II
CHARACTERISTICS OF THE EVALUATION DATASETS

Dataset	Length (bp)	Source / Organism
Dataset 1	8,000 bp	<i>Homo sapiens</i> mitochondrion (NCBI)
Dataset 2	8,000 bp	<i>Escherichia coli</i> (NCBI)

A. Dataset

Two types of biological sequence datasets were used to evaluate the proposed framework under different similarity conditions. The first dataset consists of human mitochondrial DNA sequences (**Homo sapiens mitochondrion, complete genome**) from the NCBI nucleotide database [16]. The second dataset consists of bacterial genomic sequences (**E. coli genome**) from NCBI [17], selected to represent low-similarity cases and simulate challenging evolutionary distances.

All sequences were preprocessed by removing FASTA headers and retaining only valid nucleotide characters (A, T, C,

G). Sequences were standardized to a fixed length of **8000 nucleotides** for consistency. Each experiment was repeated for 10 independent runs, and results are reported as averages to reduce variance from random execution-time fluctuations. Evaluation criteria include alignment score, execution time, and computational efficiency.

B. Alignment Score Comparison

Fig. 5 presents the alignment score comparison between the traditional SW algorithm and the proposed optimized approaches. The adaptive model achieved higher alignment scores than fixed banded alignment while maintaining computational efficiency.

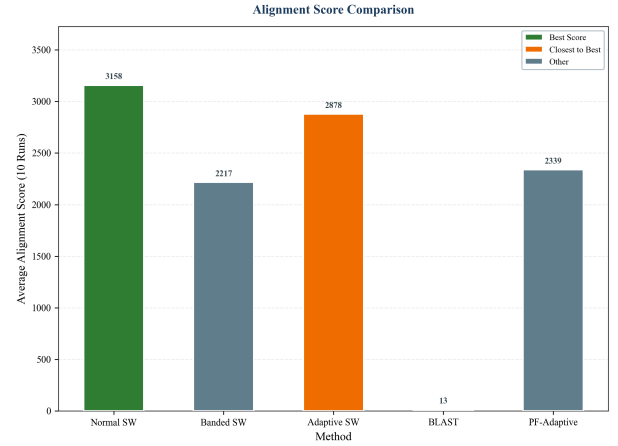


Fig. 5. Alignment score comparison among SW, banded, adaptive, BLAST, and PF-Adaptive optimization methods.

C. Scoring Matrix Heatmap

Fig. 6 illustrates the scoring matrix heatmap generated by the SW algorithm. Higher-scoring regions indicate stronger local similarities. The heatmap confirms that the adaptive approach concentrates computation around biologically relevant regions.

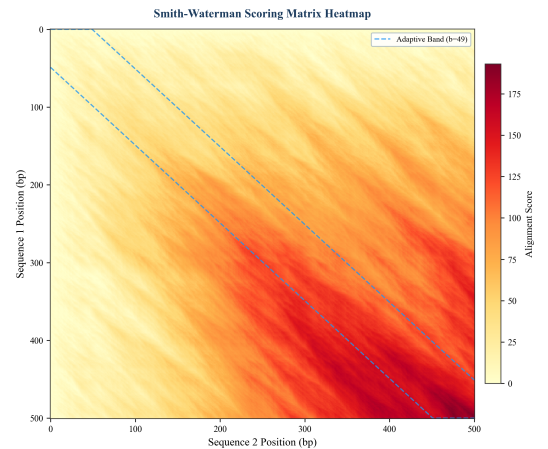


Fig. 6. Heatmap of the Smith–Waterman scoring matrix highlighting biologically significant local alignment regions.

D. Computed DP Cells

Fig. 7 compares the number of computed DP cells between the full SW and adaptive approaches. Adaptive banding significantly reduces processed cells, lowering computational overhead.

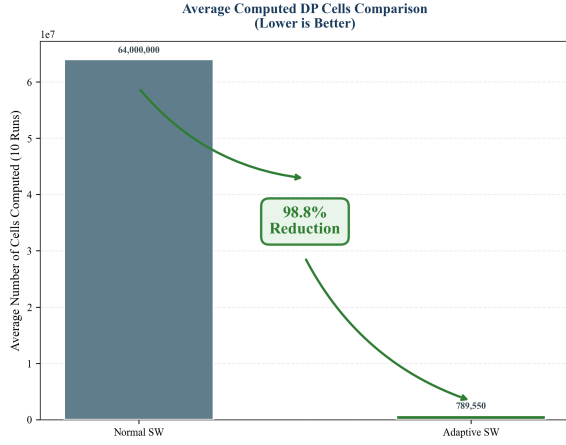


Fig. 7. Comparison of computed DP cells between full SW and adaptive alignment.

E. Similarity vs. Band Width

Fig. 8 shows the inverse relationship between sequence similarity and adaptive band width. As similarity increases, the proposed method dynamically reduces band size while preserving alignment accuracy.

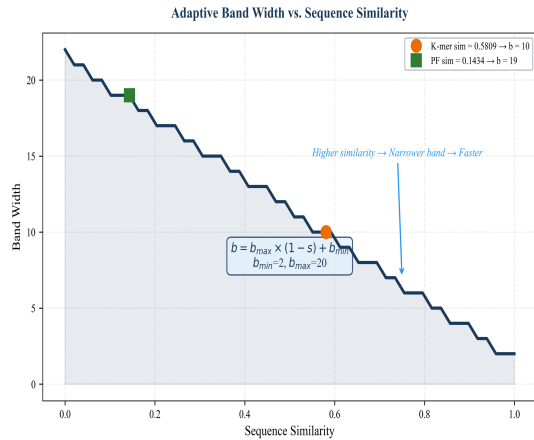


Fig. 8. Relationship between sequence similarity and adaptive band width.

F. Similarity Score Comparison

Fig. 9 compares similarity scores from different alignment approaches. The adaptive framework achieves competitive detection performance at lower computational cost than indexing-based methods.

G. Execution Time Scaling

Fig. 10 compares execution times across sequences up to 8,000 nucleotides. Standard Smith–Waterman shows an exponential runtime curve due to its full matrix traversal, whereas

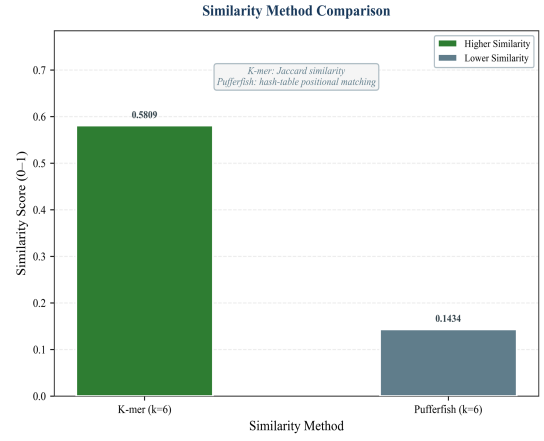


Fig. 9. Similarity score comparison between k-mer and Pufferfish-based similarity methods.

the proposed adaptive approach scales near-linearly, keeping processing times significantly lower. Table III summarizes these quantitative performance metrics, contrasting execution runtime, alignment accuracy, and calculated DP cells for each method.

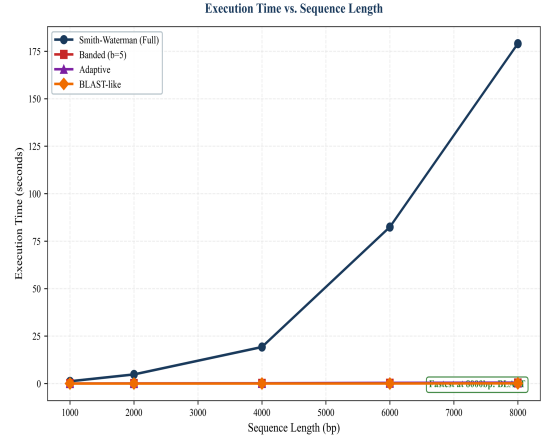


Fig. 10. Execution time scaling across different sequence lengths for SW, banded, adaptive, and BLAST methods.

TABLE III
PERFORMANCE COMPARISON OF ALIGNMENT METHODS (8000 NT SEQUENCES)

Method	Align. Score	Time (s)	DP Cells
Standard SW	3158.00	93.7443	Full matrix
Fixed Banded	2217.00	0.1531	Narrow band
Adaptive (Proposed)	2877.94	1.6495	789,550
BLAST-like	13.00	0.0518	Seed-extend

The traditional SW algorithm achieved the highest alignment score (3158.00) due to exhaustive full-matrix computation, but required the longest execution time (93.7443 s). The fixed banded approach reduced runtime to 0.1531 s but

produced a lower score of 2217.00. The proposed adaptive method achieved a score of 2877.94 with an execution time of only 1.6495 s, processing only 789,550 DP cells instead of the full matrix. The BLAST-like heuristic was the fastest (0.0518 s), but produced a drastically lower alignment score of 13, reflecting reduced sensitivity. Overall, the proposed adaptive framework provides a practical trade-off between speed and alignment quality for large-scale genomic analysis.

VI. CONCLUSION

This paper integrates the Starfish Optimization Algorithm with adaptive banding to optimize local sequence alignment. By dynamically adjusting the search corridor based on sequence similarity, the framework significantly reduces Dynamic Programming matrix computations while preserving alignment sensitivity on par with exhaustive Smith–Waterman. This approach is particularly advantageous for biological datasets with heterogeneous mutation densities. Nevertheless, the pipeline remains partially sensitive to initial k-mer similarity estimates, which may necessitate secondary heuristic adjustments when processing extremely divergent evolutionary sequences.

Future work should investigate the applicability of the proposed approach to multiple sequence alignment (MSA) or structure-based protein comparisons.

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